Problem 6. (10 points)

Suppose that $A \in \mathbb{R}^{m \times n}$, $\mathbf{b} \in \mathbb{R}^m$ and $\mathbf{x} \in \mathbb{R}^n$.
Let $f(\mathbf{x}) = \|\mathbf{b} - A\mathbf{x}\|_2^2$. Show that $\nabla f(\mathbf{x}) = 2A^\top(A\mathbf{x} - \mathbf{b})$

Answer: $A\mathbf{x} = \begin{bmatrix} \sum_j a_{1j}\mathbf{x}_j \\ \sum_j a_{2j}\mathbf{x}_j \\ \vdots \\ \sum_j a_{nj}\mathbf{x}_j \end{bmatrix}$. $A\mathbf{x} - \mathbf{b} = \begin{bmatrix} \sum_j a_{1j}\mathbf{x}_j - \mathbf{b}_j \\ \sum_j a_{2j}\mathbf{x}_j - \mathbf{b}_j \\ \vdots \\ \sum_j a_{nj}\mathbf{x}_j - \mathbf{b}_j \end{bmatrix}$ or $\sum_i \left(\sum_j a_{ij}\mathbf{x}_j - \mathbf{b}_j\right)$.
$\|A\mathbf{x} - \mathbf{b}\|_2^2 = \sum_i \left(\sum_j a_{ij}\mathbf{x}_j - \mathbf{b}_j\right)^2$.

$$\nabla f(\mathbf{x}) = \begin{bmatrix} \frac{\partial}{\partial x_1} \\ \frac{\partial}{\partial x_2} \\ \vdots \\ \frac{\partial}{\partial x_n} \end{bmatrix}$$

Lets take the derivative with respect to some element $\mathbf{x}_k$ where $1 \leq k \leq n$.

$\frac{\partial}{\partial x_k}\|A\mathbf{x} - \mathbf{b}\|_2^2 = \frac{\partial}{\partial x_k}\sum_i \left(\sum_j a_{ij}\mathbf{x}_j - \mathbf{b}_j\right)^2 = \sum_i \frac{\partial}{\partial x_k}\left(\sum_j a_{ij}\mathbf{x}_j - \mathbf{b}_j\right)^2$.

Using the chain rule, we can continue: $\sum_i 2\left(\sum_j a_{ij}\mathbf{x}_j - \mathbf{b}_j\right)\frac{\partial}{\partial x_k}\left(\sum_j a_{ij}\mathbf{x}_j - \mathbf{b}_j\right)$.

Solving $\frac{\partial}{\partial x_k}\left(\sum_j a_{ij}\mathbf{x}_j - \mathbf{b}_j\right)$ alone, we can see that the derivative will be zero for every value of $j \neq k$, so we can rewrite this as: $\frac{\partial}{\partial x_k}(a_{ik}\mathbf{x}_k - \mathbf{b}_k) = a_{ik}$.

$= \sum_i 2a_{ik}\left(\sum_j a_{ij}\mathbf{x}_j - \mathbf{b}_j\right) = 2\sum_i (A\mathbf{x} - \mathbf{b})_i a_{ik} = 2A_k^T(A\mathbf{x} - \mathbf{b})$.

Up to now, we have shown that $\frac{\partial}{\partial x_k}\|A\mathbf{x} - \mathbf{b}\|_2^2 = 2A_k^T(A\mathbf{x} - \mathbf{b})$, but we would like to show this for all values of k so that we get $\nabla f(\mathbf{x})$. Lets rewrite it below:

$$\nabla f(\mathbf{x}) = \begin{bmatrix} \frac{\partial}{\partial x_1} \\ \frac{\partial}{\partial x_2} \\ \vdots \\ \frac{\partial}{\partial x_n} \end{bmatrix} = \begin{bmatrix} 2A_1^T(A\mathbf{x} - \mathbf{b}) \\ 2A_2^T(A\mathbf{x} - \mathbf{b}) \\ \vdots \\ 2A_n^T(A\mathbf{x} - \mathbf{b}) \end{bmatrix} = 2\begin{bmatrix} A_1^T \\ A_2^T \\ \vdots \\ A_n^T \end{bmatrix}(A\mathbf{x} - \mathbf{b}) = 2A^T(A\mathbf{x} - \mathbf{b}).$$

Thus, we have shown the desired equality. ■